# Andrejs Fedjajevs

## AI \ Data Scientist

Over 6 years of experience with data science and signal processing for (contractual) research and advanced development teams, in direct contact with partners.
Strong individual contributor and algorithm development tech lead.

I enjoy diving into specialized topics, assist business in sharpening the use-case and support it with numbers.
My goal is to use sensors, data and computing to create insights and value.

### Contacts


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### Skills

Signal Processing
Machine Learning
Data Engineering
Statistics
Scientific Programming
Visualization \ Dashboards

Research
Use-case definition
Agile \ PM

Wearables \ Vital Signs
Biomedical Modeling
Sensors \ Optics

### Tools

Python, MATLAB, SQL, Java
PyTorch, xGB,
PyQt, Plotly, PowerBI,
Azure ML, Git Actions,
Jira

### Languages

English
Dutch
Latvian
Russian

### Koninklijke Philips

**Data Scientist** Apr 2022 - Present

Create measurement innovations for in-hospital technologies and make out-of-hospital care more efficient

• Developed **AI models and algorithms** for clinical telemonitoring device, improving its activity monitoring and gait analysis functionality
• Research the problem of oxygenation signal accuracy and biases; validation of alternative (non-finger) SpO2 optical sensors
• Promoted better way-of-working with algorithms and **data management practices** within the department

### Imec

**Senior R&D Engineer** Jan 2021 - Apr 2022

**R&D Engineer** Oct 2016 - Jan 2021

Data science, research and algorithms development for wearables and health-tech applications

• Developed (real-time) **algorithms and models** for heart rate, respiration, HRV, detection of stress, cardiology conditions, etc. from various sensors (ECG, PPG, BioZ, EEG, radar, camera)

• **Competence team lead** for biomedical algorithm developers. Created efficient learning environment and internal assets management. We improved data analysis work reuse and reliability within the department, promoted more agile approach within non-software-development projects.

• PM for consultancy projects and ITEA-funded consortium Inno4Health (Data Analysis&AI task). I curated activities for implementing personalized health data analytics solutions, aligned effort with stakeholders.

• Build a real-time features extraction pipeline from a portable EEG headset for **neuromarketing** application. Designed a validation framework based on **Azure MLOps** services and optimized an algorithm for noise and blink removal

• Implemented a **deep learning pipeline** for real-time human gait analysis. The model derived crucial gait parameters from one IMU sensor's raw data stream. The team delivered a prototype device to track patients' recovery after knee surgery

• Responsible for **data science** and processing pipelines for wearable (cuff-less) blood pressure device. Years long R&D work led to meeting AAMI standard requirements using in-house designed wristband module

## Education & Learning

### Master's in Electrical Engineering

**2014 - 2016**

Track: Signals and Systems

Thesis: Ultrasound Imaging Using a Single Element Transducer

### Specialist in Electrical Engineering

**2009 - 2014**

Track: Devices and Methods of Quality Control and Diagnostics

## Publications


Campero Jurado, I. et al. (2022) "Interpretable assessment of st-segment deviation in ECG Time Series," Sensors, 22(13), p. 4919.

Fedjajevs, A. et al. (2021) "Evaluation of HRV from repeated measurements of PPG and arterial blood pressure signals," 2021 Computing in Cardiology.

Fedjajevs, A. et al. (2020) "Platform for analysis and labeling of Medical Time Series," Sensors, 20(24), p. 7302.

Kruizinga, P. et al. (2017) "Compressive 3D ultrasound imaging using a single sensor," Science Advances, 3(12).